CANONICAL ISOMORPHISM OF LEFT AND RIGHT FRACTION CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4984-4993, following the statement at 4977-4982.

Editorial note. The proof is complete; both directions are covered by the two universal properties and their uniqueness clauses.

Allowed standard prerequisites (editorial).

- Universal property of the left fraction localization. lemma-properties-left-localization, cited at proof lines 4986-4988. Supplies the universal property (existence and uniqueness of factorization through Q_left) used to obtain C_left -> C_right and to prove mutual inverse-ness.
- Universal property of the right fraction localization. lemma-properties-right-localization, cited at proof lines 4986-4988. Supplies the symmetric universal property for C_right used for the reverse functor.

Source reference. Original lemma 04VG: sources/stacks-categories/pinned-original-source.tex, source lines 4623–4623. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04VK: sources/stacks-categories/pinned-original-source.tex, source lines 4910–4910. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category and let S be a multiplicative system. The category of left fractions and the category of right fractions $S^{-1}\mathcal{C}$ are canonically isomorphic.*

Proof. Denote $\mathcal{C}_{left}$, $\mathcal{C}_{right}$ the two categories of fractions. By the universal properties of Lemmas 04VG and 04VK we obtain functors $\mathcal{C}_{left} \to \mathcal{C}_{right}$ and $\mathcal{C}_{right} \to \mathcal{C}_{left}$. By the uniqueness statement in the universal properties, these functors are each other's inverse. □

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.